



Answers and Solutions

1. D. 1/3

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\frac{1}{2} = \frac{1}{4} + \frac{1}{3} - P(A \cap B)$$

$$P(A \cap B) = \frac{1}{12}$$

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{1/12}{1/4} = \frac{1}{3}$$

2. A. 1/2

$$P(\text{snow}_M | \text{snow}_S) = \frac{P(\text{snow}_M \& \text{snow}_S)}{P(\text{snow}_S)}$$

$$= \frac{\frac{3}{10}}{\frac{3}{5}} = \frac{1}{2} \rightarrow \text{Ans}$$

3. B. 60

$$N = {}_5C_2 \cdot {}_4C_2 = 60 \rightarrow \text{Ans}$$

4. C. 0.000129

$$P = \left(\frac{1}{6}\right)^5 = 0.000129 \rightarrow \text{Ans}$$

5. A. 0.000625

$$P = \left(\frac{5}{200}\right)\left(\frac{5}{200}\right) = 0.000625$$

6. D. 5 %

$$P = \frac{5}{100} = 0.05 = 5\% \rightarrow \text{Ans}$$

7. A. 0.950625

$$P(\text{first is non-def} \cap \text{second is non-def})$$

$$= \left(\frac{195}{200}\right)\left(\frac{195}{200}\right) = 0.950625$$

8. D. 25 %

Let:

A = event that a part has a type 2 defect

B = event that a part has also a type 1 defect

$$P(B|A) = \frac{P(A \cap B)}{P(B)} = \frac{0.005}{0.02} = 0.25$$

9. A. 2,986,960

$${}_{52}C_5 = \frac{52!}{5!(52-5)!} = 2,986,960$$

10. D. 5/14

P_1 = probability of drawing a black ball in the first draw

P_2 = probability of drawing a black ball in the second draw

$$P_1 = \frac{5}{8} \quad P_2 = \frac{4}{7}$$

Note:

The 1st ball was not returned in the bag before the 2nd ball was drawn

P = probability that both balls drawn are all black

$$P = P_1 \times P_2 = \frac{5}{8} \times \frac{4}{7} = \frac{20}{56} = \frac{5}{14}$$

11. C. 1/9

Total number of trials = 36

Number of trials with a sum of 10 or 12 = 4

	1	2	3	4	5	6
1						
2						
3						
4						
5						
6						

$$P = \frac{\text{number of successful trials}}{\text{total number trials}}$$

$$P = \frac{4}{36} = \frac{1}{9}$$

12. A. 64

$$N = {}_4P_1 + {}_4P_2 + {}_4P_3 + {}_4P_4$$

$$N = \frac{4!}{(4-1)!} + \frac{4!}{(4-2)!} + \frac{4!}{(4-3)!} + \frac{4!}{(4-4)!}$$

$$N = 64 \text{ signals}$$

13. A. 0.3125

Binomial Distribution ${}^nC_r (p)^r (q)^{n-r}$
 $n = 5$ times $r = 3$ times

$$p = \frac{4}{6} = \frac{2}{3}$$

$$P = {}_5C_3 \left(\frac{1}{2}\right)^3 \left(1 - \frac{1}{2}\right)^{5-3} = 0.3125$$

14. A. 0.7

$$P = 1 - P(A \text{ or } B)$$

$$P = 1 - [P(A) + P(B) - P(A \text{ and } B)]$$

$$P = 1 - [0.25 + 0.15 - 0.1] = 0.7$$

15. C. 0.533

$$P = \left(\frac{4}{10}\right)\left(\frac{6}{9}\right) + \left(\frac{6}{10}\right)\left(\frac{4}{9}\right) = \frac{8}{15} = 0.533$$

16. C. 360,360

$$N = (15)(14)(13)(12)(11) = 360,360$$

17. B. 1680

$$N = \frac{n!}{p!q!r!}$$

n = number of total samples

p, q, r = identical samples

$$N = \frac{8!}{3!2!2!} = 1680$$

18. D. Sunbathing all day and getting a sunburn

19. B. 0.7265

$$P = \left(\frac{90}{100}\right)\left(\frac{89}{99}\right)\left(\frac{88}{98}\right) = 0.7265$$

20. A. 100,800

$$N = ({}_6C_2)({}_8C_3)(5!) = 100,800$$

21. B. 0.8077

$$P = \frac{(21)(25)(24)}{(26)(25)(24)} = 0.8077$$

22. C. 80/243



$$P = {}_5C_3 \left(\frac{2}{3}\right)^3 \left(1 - \frac{2}{3}\right)^{5-3} = \frac{80}{243}$$

23. A. 2/3

2 heads, 1 tail

Desired outcome: HHT, HTH

Total possible outcomes: HHT, HTH, THH

$$P = \frac{2}{3}$$

24. A. 13,860

$$N = \frac{n!}{p!q!r!} = \frac{12!}{2!4!6!} = 13,860$$

25. D. Tossing two coins and getting two heads or two tails.

26. A. 0.857

Using Bayes' Theorem:

$$P(G|E) = \frac{P(E|G) \cdot P(G)}{P(E|G) \cdot P(G) + P(E|T) \cdot P(T)}$$

$$P(G|E) = \frac{(0.20)(0.75)}{(0.20)(0.75) + (0.10)(0.25)}$$

$$P(G|E) = 0.857$$

27. A. 0.375

For 3-element word that takes the value of 1 or 2, the possible outcomes are:

111 121 122 221 211 222

There are a total of 8 outcomes. The words that contain exactly two elements of value 1 are:

112 121 211

There are a total of 3 words that contain exactly two elements of value 1. Therefore:

$$P = \frac{3}{8} = 0.375$$

28. D. Drawing a card from a deck and getting a club and an ace

29. D. 1/3

$P(B|A)$ = probability that B will happen given that A happened.

$P(A|B)$ = probability that A will happen given that B happened. So,

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

$$\frac{2}{3} = \frac{P(A \cap B)}{1/4}$$

$$P(A \cap B) = \frac{1}{6}$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$\frac{1}{2} = \frac{1/6}{P(B)}$$

$$P(B) = \frac{1}{3}$$

30. B. 0.3

$$P = 1 - P(A \text{ or } B)$$

$$P = 1 - [P(A) + P(B) - P(A \text{ and } B)]$$

$$P = 1 - [0.6 + 0.3 - 0.2] = 0.3$$

31. B. 34,650

$$N = \frac{11!}{(4!)(4!)(2!)} = 34,650$$

32. B. 120

$$N = (6 - 1)! = 120$$

33. B. 1/16

Binomial Distribution:

$$P = {}_nC_r p^r q^{n-r}$$

$$\text{Probability of success (tail), } p = \frac{1}{2}$$

$$\text{Probability of failure (head), } q = \frac{1}{2}$$

$$\text{Total coins, } n = 7$$

Total coins with tails as a result, $r = 6, 7$

$$P = {}_7C_6 \left(\frac{1}{2}\right)^6 \left(\frac{1}{2}\right)^{7-6} + {}_7C_7 \left(\frac{1}{2}\right)^7 \left(\frac{1}{2}\right)^{7-7}$$

$$P = \frac{1}{16}$$

34. B. 0.176

$$P = {}_nC_r p^r q^{n-r} = {}_{20}C_{10} \left(\frac{1}{2}\right)^{10} \left(\frac{1}{2}\right)^{20-10}$$

$$P = 0.176$$

35. D. 0.215

$$P = {}_nC_r p^r q^{n-r} = {}_{10}C_7 (0.60)^7 (0.40)^{10-7}$$

$$P = 0.215$$

36. C. 560

Permutation with identical elements: Considering the targets per column are identical,

$$P = \frac{n!}{p!q!s!} = \frac{8!}{3!3!2!} = 560$$

37. D. 0.507

Probability that they have different birthdays, P_1 :

$$P_1 = \frac{{}^{365}P_{23}}{365^{23}} = 0.493$$

Probability that at least two of them have the same birthday, P :

$$P = 1 - 0.493 = 0.507$$

38. D. 40/81

Assume the first draw is black and the second draw is white:

$$P_1 = P_{\text{black}} \times P_{\text{white}} = \frac{4}{9} \times \frac{5}{9} = \frac{20}{81}$$

Assume the first draw is white and the second draw is black:

$$P_2 = P_{\text{white}} \times P_{\text{black}} = \frac{5}{9} \times \frac{4}{9} = \frac{20}{81}$$

Let: P = probability that one ball is black and the other is white

$$P = P_1 + P_2 = \frac{20}{81} + \frac{20}{81} = \frac{40}{81}$$

39. D. 89.6 %

Probability of getting a passing score in each subject is 0.8

Probability of failing in any of the three subjects is 0.2



Let: P_1 = probability of passing exactly two subjects

$$P_1 = {}_3C_2 (0.8)^2 (0.2)^1$$

$$P_1 = \frac{3!}{(3-2)!2!} (0.128) = 0.384$$

P_2 = probability of passing all the three subjects

$$P_2 = {}_3C_3 (0.8)^3 (0.2)^0$$

$$P_2 = (1)(0.512) = 0.512$$

P = probability of passing at least two subjects

$$P = 0.384 + 0.512 = 0.896$$

40. B. 1/4

$$P = {}_nC_r p^r q^{n-r}$$

where: $p = \frac{1}{2}$; $q = \frac{1}{2}$; $n = 3$; $r = 3$

$$P_{3H} = {}_3C_3 \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^0 = (1) \left(\frac{1}{2}\right)^3 = \frac{1}{8}$$

$$P_{3T} = {}_3C_3 \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^0 = (1) \left(\frac{1}{2}\right)^3 = \frac{1}{8}$$

$$P = P_{3H} + P_{3T} = \frac{1}{8} + \frac{1}{8} = \frac{1}{4}$$

41. D. 1365

$$N = {}_{15}C_4 = \frac{15!}{(15-4)!4!} = 1,365$$

42. C. 60

Number of A's = 3

Number of N's = 2

$$N = \frac{n!}{p!q!} = \frac{6!}{3!2!} = 60 \text{ ways}$$

43. B. 4/13

In a pack of 52 playing cards, there are 4 king cards, 13 diamond cards and 1 king & diamond card at the same time

P_K = probability of drawing a king

P_D = probability of drawing a diamond

$P_{K \& D}$ = probability of drawing a king at the same time a diamond

$$P_{K \text{ or } D} = P_K + P_D - P_{K \& D}$$

$$P_{K \text{ or } D} = \frac{4}{52} + \frac{13}{52} - \frac{1}{52}$$

$$P_{K \text{ or } D} = \frac{16}{52} = \frac{4}{13}$$

44. B. 0.53

Assume the first draw is black and the second draw is white:

$$P_1 = P_{\text{black}} \times P_{\text{white}} = \frac{4}{10} \times \frac{6}{9} = \frac{24}{90}$$

Assume the first draw is white and the second draw is black:

$$P_2 = P_{\text{white}} \times P_{\text{black}} = \frac{6}{10} \times \frac{4}{9} = \frac{24}{90}$$

Let: P = probability that one ball is black and the other is white

$$P = P_1 + P_2 = \frac{24}{90} + \frac{24}{90} = 0.533$$

45. A. 1.9 %

$$P(D) = P(D \cap S1) + P(D \cap S2) + P(D \cap S3)$$

$$P(D) = P(S1)P(D | S1) + P(S2)P(D | S2) + P(S3)P(D | S3)$$

$$P(D) = (0.3)(0.01) + (0.5)(0.02) + (0.2)(0.03)$$

$$P(D) = 0.019 = 1.9\%$$

46. D. 0.093

$$P(\text{honest}) = 0.12$$

$$P(\text{dishonest}) = 1 - 0.12 = 0.88$$

$$P = (0.88)(0.88)(0.12)$$

$$P = 0.092928$$

47. C. 0.329

$$P = {}_nC_r p^r q^{n-r}$$

$$n = 6, r = 2, p = \frac{1}{3}, q = 1 - \frac{1}{3} = \frac{2}{3}$$

$$P = {}_6C_2 \left(\frac{1}{3}\right)^2 \left(\frac{2}{3}\right)^{6-2} = 0.329 \rightarrow \text{Ans}$$

48. B. 8/15

The probability that bus is on time and Mr. Alarde is driver is:

$$P = \frac{2}{3} \left(\frac{4}{5}\right) = \frac{8}{15} \rightarrow \text{Ans}$$

49. 3/20

$$P(\text{hit}_{\text{Selena}}) = \frac{8}{20}$$

$$P(\text{hit}_{\text{Tracey}}) = \frac{6}{16}$$

$$P(\text{hit}) = \frac{8}{20} \left(\frac{6}{16}\right) = \frac{48}{320} \text{ or } \frac{3}{20} \rightarrow \text{Ans}$$

50. D. 37.5%

A: seen an advertisement for the new soap

B: tried the new soap

$$P_A = 0.4 ; P_B = 0.20$$

$$P_{A \cap B} = 0.15$$

Conditional Probability:

Probability of B event given A event

$$P_{B|A} = \frac{P_{B \cap A}}{P_A}$$

$$P_{B|A} = \frac{0.15}{0.4} \times 100$$

$$P_{B|A} = 37.5\% \rightarrow \text{Ans}$$